Calculus II

Tutorial #11 28 Apr - 02 May 2025

Topics:

- Some simple forms of second-order differential equations
- Wronskian
- Homogeneous linear differential equations with constant-coefficients
- The method of undetermined coefficients

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. Let's consider a second-order differential equation

$$\frac{d^2y}{dx^2} = F\left(x, y, \frac{dy}{dx}\right),$$

for which there are two cases: (1) the dependent variable y is missing, and (2) the independent variable x is missing.

   a) If case (1) happens, then a change of variables $z = \dfrac{dy}{dx}$ reduces the equation to a first-order differential equation w.r.t. z and x.

   b) If case (2) happens, then a change of variables $z = \dfrac{dy}{dx}$ reduces the equation to a first-order differential equation w.r.t. z and y.

   c) In both cases (1) and (2), a chain rule of differentiation is applied.

2. The functions $f_1(x) = x$, $f_2(x) = x + x^2$, and $f_3(x) = 2x - x^2$ are linearly dependent.

3. If the Wronskian of a set of functions is identically zero at every point of an interval I, then the set of functions is linearly dependent.

• **Problems**.

4. Solve the following differential equations:

   (a) $y'' = 2x^{-1}y' + 4x^2$

   (b) $(x-1)(x-2)y'' = y' - 1$

5. Solve the following differential equations:

   (a) $y'' + 2y^{-1}(y')^2 = y'$

   (b) $y'' = (y')^2 \tan y$.

6. For all constants a and b, prove that the set of functions

$$\{e^{ax}\cos bx, e^{ax}\sin bx, xe^{ax}\cos bx, xe^{ax}\sin bx\}$$

   is linearly independent on $(-\infty, \infty)$.

7. Find the general solution to the following homogeneous differential equation.

   (a) $y'' - 5y' + 6y = 0$.

   (b) $y'' + 10y' + 25y = 0$.

   (c) $y'' + 8y' + 20y = 0$.

In Problems 8–10, using the method of undetermined coefficients, find the general solution to the given differential equation.

8. $$y'' + 4y' + 4y = e^{-x}\cos x.$$

9. $$y'' + y = 6e^{x}.$$

10. $$y'' + 16y = 4\cos x.$$

END

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Answers to selected problems

1. a) True b) True c) False

2. True

3. False

4. (a) $y = x^4 + C_1x^3 + C_2$ (b) $y = \log|x-1| + C_1x - C_1\log|x-1| + C_2$

5. (a) $y = \sqrt[3]{C_1 + C_2e^x}$ (b) $y = \sin^{-1}(C_1x + C_2)$

7. (a) $y(x) = C_1e^{2x} + C_2e^{3x}$ (b) $y(x) = e^{-5x}(C_1 + C_2x)$ (c) $y(x) = e^{-4x}(C_1\cos 2x + C_2\sin 2x)$

8. $y(x) = C_1 e^{-2x} + C_2 x e^{-2x} + \dfrac{1}{2} e^{-x} \sin x$
9. $y(x) = C_1 \cos x + C_2 \sin x + 3e^{x}$
10. $y(x) = C_1 \cos 4x + C_2 \sin 4x + \frac{4}{15} \cos x.$